

Math 1232 Summer 2026

Practice Midterm

- You will have 90 minutes for this test.
- You are not allowed to consult books or notes during the test, but you may use a one-page, one-sided cheat sheet you have made for yourself ahead of time.
- You may use a four-function calculator. You may leave answers unsimplified, except you should compute trigonometric functions as far as possible.
- The exam has 10 problems, and is 6 pages total.
- Each problem is worth ten points. The whole test is scored out of 100 points.
- Read the questions carefully and make sure to answer the actual question asked. Make sure to justify your answers—math is largely about clear communication and argument, so an unjustified answer is much like no answer at all. When in doubt, show more work and write complete sentences.
- If you need more paper to show work, I have extra at the front of the room.
- Good luck!

1	2	3	4	5
6	7	8	9	10

Total /100

Name: **Solutions**

Problem 1. Compute $\int_1^2 \frac{e^{1/x}}{x^2} dx$.

Solution. We take $u = 1/x$ so $du = \frac{-1}{x^2} dx$. Then

$$\begin{aligned}\int_1^2 \frac{e^{1/x}}{x^2} dx &= \int_1^{1/2} -e^u du \\ &= -e^u \Big|_1^{1/2} = -e^{1/2} + e^1 = e - \sqrt{e}.\end{aligned}$$

Problem 2. Compute $\frac{d}{dx}(\sin(x))^{\arcsin(x)}$.

Solution.

$$\begin{aligned}\ln |y| &= \arcsin(x) \ln |\sin(x)| \\ \frac{y'}{y} &= \frac{\ln |\sin(x)|}{\sqrt{1-x^2}} + \frac{(\cos(x)) \arcsin(x)}{\sin(x)} \\ y' &= y \left(\frac{\ln |\sin(x)|}{\sqrt{1-x^2}} + \cot(x) \arcsin(x) \right) \\ &= \sin(x)^{\arcsin(x)} \left(\frac{\ln |\sin(x)|}{\sqrt{1-x^2}} + \cot(x) \arcsin(x) \right).\end{aligned}$$

Problem 3. Let $h(x) = \sqrt{x^5 + x + 2}$. Compute $(h^{-1})'(6)$

Solution. By the Inverse Function Theorem, we know that

$$(h^{-1})'(2) = \frac{1}{h'(h^{-1}(6))}.$$

Guess and check shows that $h(2) = 6$, so $h^{-1}(6) = 2$. And we know that

$$h'(x) = \frac{1}{2}(x^5 + x + 2)^{-1/2}(5x^4 + 1)$$

and thus

$$h'(2) = \frac{1}{12}(81)$$

Therefore

$$(h^{-1})'(6) = \frac{12}{81}.$$

Problem 4. Compute $\lim_{x \rightarrow 1} (2x - 1)^{\frac{1}{\ln(x)}}$.

Solution. We are looking at an indeterminate form of type 1^∞ , so we should take L to be the limit and try to use log rules to compute $\ln(L)$ instead. We have

$$\ln(L) = \lim_{x \rightarrow 1} \frac{1}{\ln(x)} \ln(2x - 1) = \lim_{x \rightarrow 1} \frac{\ln(2x - 1)}{\ln(x)},$$

which gives an indeterminate form of type $0/0$. Thus we can apply l'Hôpital's rule to get

$$\ln(L) = \lim_{x \rightarrow 1} \frac{\frac{2}{2x-1}}{\frac{1}{x}} = \lim_{x \rightarrow 1} \frac{2x}{2x-1} = \frac{2}{1} = 2.$$

Now $\ln(L) = 2$ implies $L = e^2$.

Problem 5. Compute $\int \frac{1}{x\sqrt{1+x^2}} dx$.

Solution. We see a $\sqrt{1+x^2}$ in the integrand, so we should try the trig substitution $x = \tan(\theta)$. This gives $dx = \sec^2(\theta) d\theta$, and so we can plug everything in and get

$$\begin{aligned} \int \frac{1}{x\sqrt{1+x^2}} dx &= \int \frac{1}{\tan(\theta)\sqrt{1+\tan^2(\theta)}} \sec^2(\theta) d\theta \\ &= \int \frac{\sec^2 \theta}{\tan(\theta)\sqrt{\sec^2(\theta)}} d\theta = \int \frac{\sec^2(\theta)}{\tan(\theta)\sec(\theta)} d\theta \\ &= \int \frac{\sec(\theta)}{\tan(\theta)} d\theta = \int \frac{\frac{1}{\cos(\theta)}}{\frac{\sin(\theta)}{\cos(\theta)}} d\theta = \int \frac{1}{\sin(\theta)} d\theta = \int \csc(\theta) d\theta. \end{aligned}$$

At this point we need an antiderivative of $\csc(\theta)$, which (along with antiderivatives of the other trig functions) is a formula we should have written down: $\int \csc(\theta) d\theta = \ln |\csc(\theta) - \cot \theta| + C$. Having done the integration, all that's left is converting back. We set $x = \tan(\theta)$, so we draw a right triangle with side x opposite angle θ and side 1 adjacent to θ . Then the missing side is $\sqrt{1+x^2}$, and we can compute

$$\csc(\theta) = \frac{\text{hyp}}{\text{opp}} = \frac{\sqrt{1+x^2}}{x} \quad \text{and} \quad \cot(\theta) = \frac{\text{adj}}{\text{opp}} = \frac{1}{x}.$$

Thus

$$\int \frac{1}{x\sqrt{1+x^2}} dx = \ln |\csc(\theta) - \cot(\theta)| + C = \ln \left| \frac{\sqrt{1+x^2}}{x} - \frac{1}{x} \right| + C.$$

Problem 6. Compute $\int \frac{x+11}{(x-2)(x^2+9)} dx$.

Solution. We need to do a partial fractions decomposition. We have

$$\begin{aligned}\frac{x+11}{(x-2)(x^2+9)} &= \frac{A}{x-2} + \frac{Bx+C}{x^2+9} \\ x+11 &= A(x^2+9) + (Bx+C)(x-2). \\ x=2: \quad 13 &= A \cdot 13 \\ A &= 1 \\ x=0: \quad 11 &= A \cdot 9 + (C)(-2) = 9 - 2C \\ 2 &= -2C \implies C = -1 \\ x=1: \quad 12 &= A \cdot 10 + (B+C)(-1) = 10 - B + 1 \\ B &= -1.\end{aligned}$$

Thus we compute

$$\begin{aligned}\int \frac{x+11}{(x-2)(x^2+9)} dx &= \int \frac{1}{x-2} + \frac{-x-1}{x^2+9} dx \\ &= \int \frac{1}{x-2} - \frac{x}{x^2+9} - \frac{1}{x^2+9} dx \\ &= \ln|x-2| - \frac{1}{2} \ln|x^2+9| - \frac{1}{3} \arctan\left(\frac{x}{3}\right) + C\end{aligned}$$

where the first and second terms are computed with u -substitutions $u = x-2$ and $u = x^2+9$ respectively (the last of which gives $du = 2x dx$, hence the factor of $1/2$), and the last term comes from the general formula

$$\int \frac{1}{x^2+a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C.$$

Problem 7. Compute $\int \sec^3(x) \tan^5(x) dx$.

Solution. With products of secant and tangent, we look for an odd power of $\tan(x)$ or an even power of $\sec(x)$ to rewrite. Here $\tan^5(x) = (\tan^2(x))^2 \tan(x) = (\sec^2(x) - 1)^2 \tan(x)$ to make the substitution $u = \sec(x)$ with $du = \sec(x) \tan(x) dx$ work out:

$$\begin{aligned}\int \sec^3(x) \tan^5(x) dx &= \int \sec^3(x) (\sec^2(x) - 1)^2 \tan(x) dx \\ &= \int u^2 (u^2 - 1)^2 du = \int u^2 (u^4 - 2u^2 + 1) du \\ &= \int u^6 - 2u^4 + u^2 du = \frac{u^7}{7} - \frac{2u^5}{5} + \frac{u^3}{3} + C \\ &= \frac{\sec^7(x)}{7} - \frac{2\sec^5(x)}{5} + \frac{\sec^3(x)}{3} + C.\end{aligned}$$

Problem 8. Compute $\int x^5 \ln(x) dx$.

Solution. We have a product of two functions, and none of our other advanced integration techniques apply, so we integrate by parts. Since $\ln(x)$ is trickier to take an antiderivative of, we don't want it to be dv , so we take $u = \ln(x)$ and $dv = x^5$ to get $du = 1/x$ and $v = x^6/6$. The integration by parts formula then gives

$$\begin{aligned}\int x^5 \ln(x) dx &= \frac{x^6}{6} \ln(x) - \int \frac{x^6}{6} \cdot \frac{1}{x} dx \\ &= \frac{x^6}{6} \ln(x) - \frac{1}{6} \int x^5 dx \\ &= \frac{x^6}{6} \ln(x) - \frac{x^6}{36} + C.\end{aligned}$$

Problem 9. Compute $\int_1^3 \frac{x}{x^2-1} dx$. Does this converge or diverge?

Solution. The integrand has a discontinuity at $x = 1$, so we compute the improper integral by letting the lower bound approach 1 from above. To find an antiderivative, we take $u = x^2 - 1$ for $du = 2x dx$.

$$\begin{aligned}\int_1^3 \frac{x}{x^2-1} dx &= \lim_{t \rightarrow 1^+} \int_t^3 \frac{x}{x^2-1} dx = \lim_{t \rightarrow 1^+} \frac{1}{2} \int_{t^2-1}^8 \frac{1}{u} du \\ &= \lim_{t \rightarrow 1^+} \frac{1}{2} \ln |u| \Big|_{t^2-1}^8 = \frac{1}{2} \ln(8) - \lim_{t \rightarrow 1^+} \ln |t^2-1| = \infty,\end{aligned}$$

since the inside of $\ln()$ is approaching zero from above, and $\ln(x)$ approaches $-\infty$ as x approaches 0 from above. Thus the integral diverges.

Problem 10. Compute the area of the surface obtained by taking the curve $y = \sqrt{15-x}$ as x goes from 3 to 5 and rotating it around the x -axis.

Solution. We have $y' = \frac{-1}{2\sqrt{15-x}}$. So we get

$$\begin{aligned}A &= \int_3^5 2\pi y \sqrt{1+y'^2} dx \\ &= \int_3^5 2\pi \sqrt{15-x} \sqrt{1 + \frac{1}{4(15-x)}} dx \\ &= 2\pi \int_3^5 \sqrt{15-x + \frac{1}{4}} dx = 2\pi \int_3^5 \sqrt{\frac{61}{4} - x} dx \\ &= -2\pi \cdot \frac{2}{3} \left(\frac{61}{4} - x \right)^{3/2} \Big|_3^5 \\ &= -\frac{4\pi}{3} \left(\left(\frac{41}{4} \right)^{3/2} - \left(\frac{49}{4} \right)^{3/2} \right).\end{aligned}$$